Unit 5, Lesson 2: Plotting on the Coordinate Plane

Benchmark

MA.6.GR.1.1 Extend previous understanding of the coordinate plane to plot rational number ordered pairs in all four quadrants and on both axes. Identify the x- or y-axis as the line of reflection when two ordered pairs have an opposite x- and y-coordinate.

Learning Targets

- ☐ I can plot rational ordered pairs.
- ☐ I can plot integers using different scales.

5.2.1 Warm-Up: Interior Designer

After getting a degree in graphic design, Soledad Alzaga built her career and company as an interior designer. Interior designers make interior spaces functional, safe, and beautiful by determining space requirements and selecting decorative items such as colors, lighting, and materials. Beyond being creative and having good taste, Soledad says to own a successful interior design business, it is necessary to have the ability to work and communicate with others, be organized, have business knowledge, and to have math and measurement skills.

1. What is something that interests you that you would like to spend your career doing?

2. What skills do you think are needed to be successful in that career?

5.2.2 Refresher: Key Terms with the Coordinate Plane

Several key terms related to the coordinate plane were used throughout Lesson 1. Below, you can find definitions for each of these key terms.

The ***coordinate plane*** is a plane determined by two perpendicular number lines, called axes. The axes intersect at the origin. Each point in the coordinate plane is represented by a pair of coordinates that represent the direction and distance from each axis. The origin is represented by the coordinate pair, $(0, 0)$.

The ***axes*** of a graph are the horizontal and vertical number lines used in a coordinate plane system.

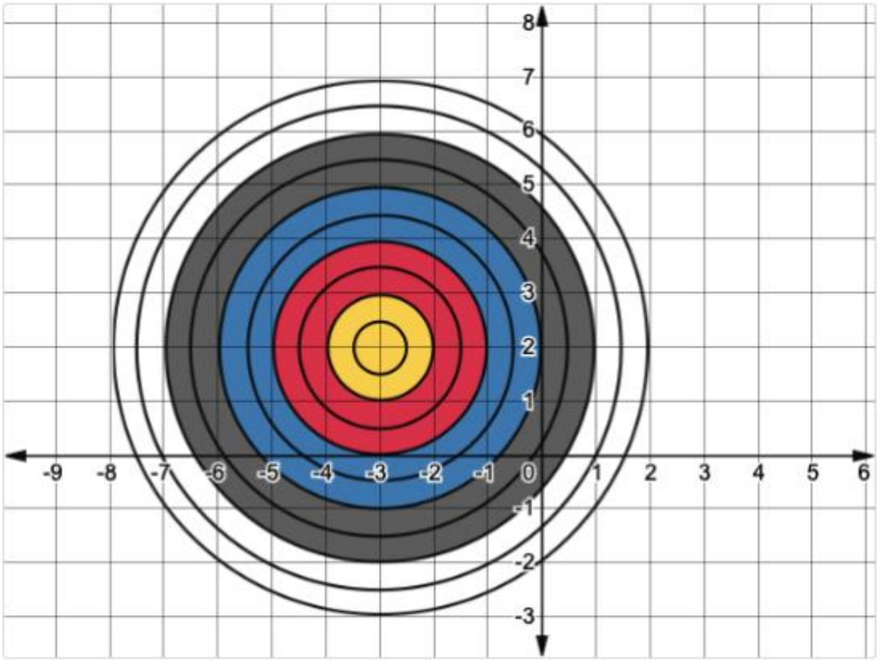
A ***coordinate*** is a number used to locate a point on a number line. A coordinate is one of the numbers in an ordered pair that locates a point on a coordinate plane.

The ***origin*** is the point of intersection of the x- and y-axes in a rectangular coordinate system, where the x-coordinate and y-coordinate are both 0.

A ***quadrant*** is any of the four regions separated by the axes in a coordinate plane.

5.2.3 Exploration: Plotting Ordered Pairs

Two players engage in a game of darts.



1. The first player throws a dart that lands at a point with an x-value halfway between -2 and -3 in the blue area. Without looking at your partner's paper, draw a point on the dart board where the dart may have landed.

    a. What are the approximate coordinates of the dart?

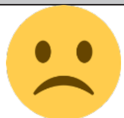
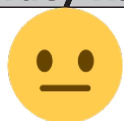
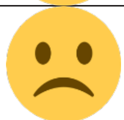
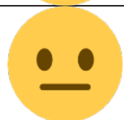
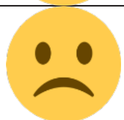
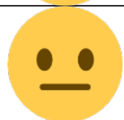
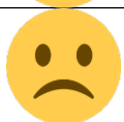
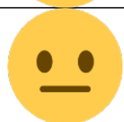
    b. What are the approximate coordinates of your partner's dart?

2. The second player throws a dart that lands in the red area, but not on one of the black circles. Without looking at your partner's paper, draw a point on the dart board where the dart may have landed.

   a. What are the approximate coordinates of the dart?

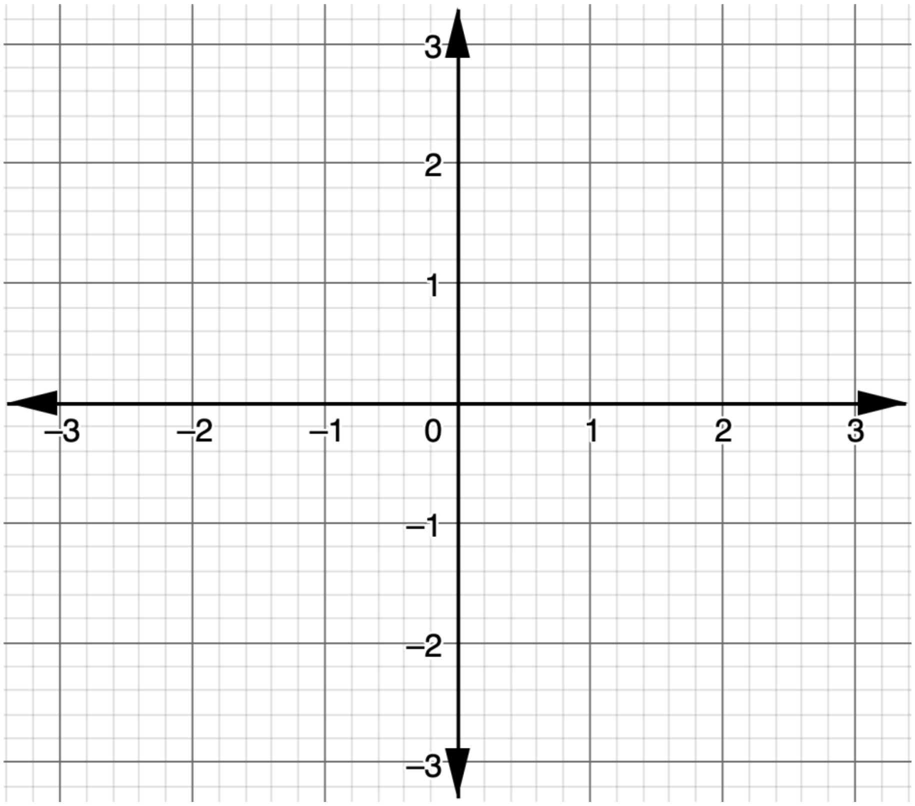
   b. What are the approximate coordinates of your partner's dart?

3. The scale used on the grid is 1 unit. Discuss with your partner how the scale used on the grid could be changed to make coordinates of the darts more accurate. Give each scale in the table a rating by circling the face that best describes the scale's accuracy, where  means the location of the coordinates would not be very accurate, and  means the location of the coordinates would be very accurate.

Scale	Accuracy Rating		
2			
1 current scale			
$\frac{1}{2}$			
$\frac{1}{4}$			

5.2.4 Guided Practice: Plotting Rational Ordered Pairs

1. On the coordinate plane, plot and label the points given.



Point A $(-2, -1)$	Point B $(1, 0.6)$	Point C $\left(\frac{3}{5}, -1\frac{2}{5}\right)$
Point D $(-0.8, -2)$	Point E $\left(-\frac{11}{5}, 1\frac{8}{10}\right)$	Point F $(2.5, 2.2)$

5.2.5 Guided Instruction: Choosing the Right Scale

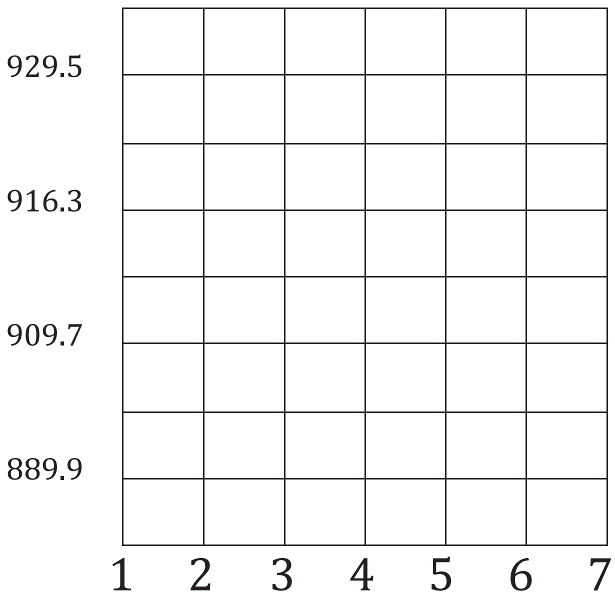
When representing data, the scale of the grid is important to effectively and accurately communicate information.

A ***scale*** is the numeric values, set at fixed intervals, assigned to the axes of a graph.

1. The table shows the total Florida tourism industry employees, in thousands, for years since 2002.

Years Since 2002	Total Tourism Industry Employees (in thousands)
1	889.9
4	909.7
5	916.3
7	929.5

To plot the data, Andrea drew the coordinate grid shown.

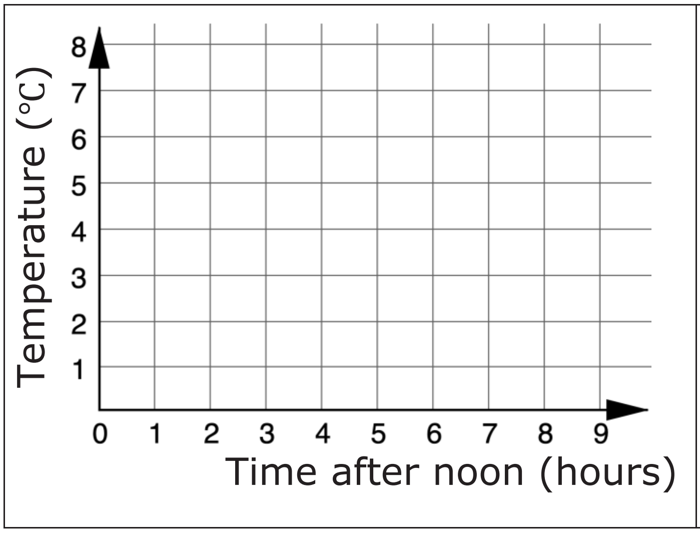


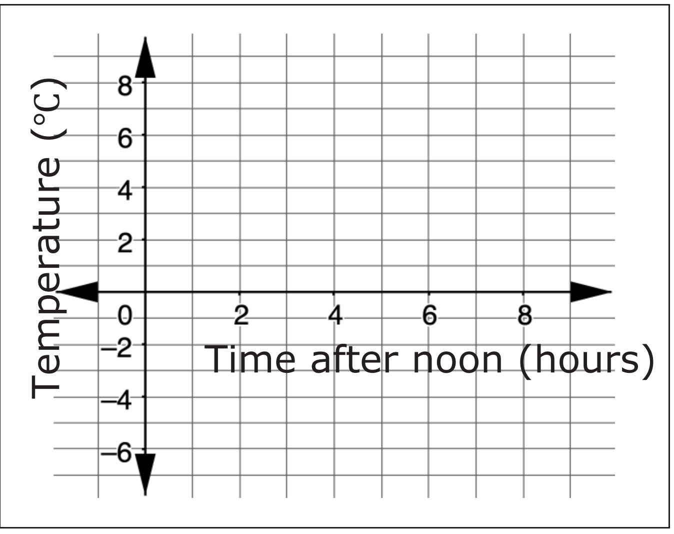
Discuss with your partner why Andrea's coordinate grid would not effectively nor accurately communicate the information about Florida tourism industry employees.

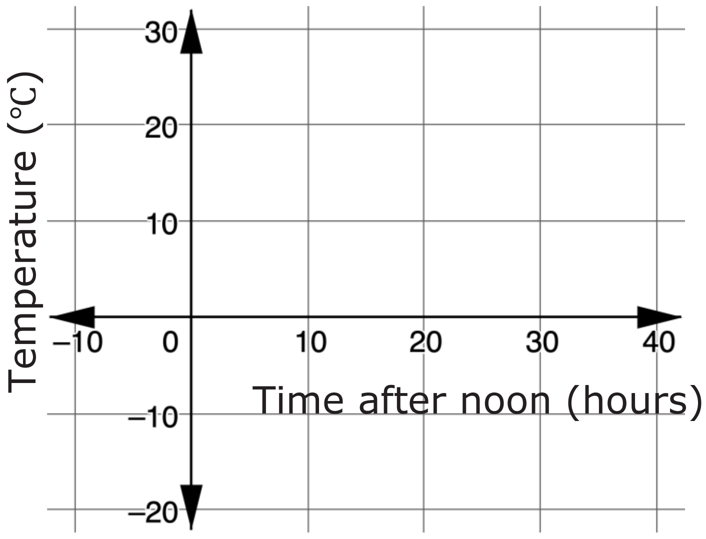
2. The data shown in the table are average autumn temperatures, in degrees Celsius (℃), in Carmathen, Wales.
3.

Time After Noon (hours)	**Temperature** (℃)
0	5
1	4
2	3
3	2
4	1
5	-2
6	-3
7	-4
8	-4

Three coordinate grids are shown, each using different scales.







a. Explain which coordinate grid most accurately represents the data.

b. Explain why the other two coordinate grids may not be as accurate.

When creating an appropriate graph, it is helpful to consider the following.

- Determine which quadrants are needed by identifying the smallest and largest points of each variable.
- Use these values to determine a good scale for the axis that represents the variable.
- Then, determine the best scale for the grid lines by considering the numbers presented in the data.
- Remember that the scale for each axis can be different, but that each axis must have a consistent scale.

4. Four coordinates are given.

$$\left(1, 2\frac{1}{2}\right) \ \left(3\frac{1}{2}, -4\right) \ (-5, -2) \ (0, 3)$$

a. What is the smallest x-coordinate?

b. What is the largest x-coordinate?

c. What is the smallest y-coordinate?

d. What is the largest y-coordinate?

e. Which quadrants need to be represented on the grid?

f. By looking at the numbers that make up the ordered pairs, what scale should be used to number each axis?

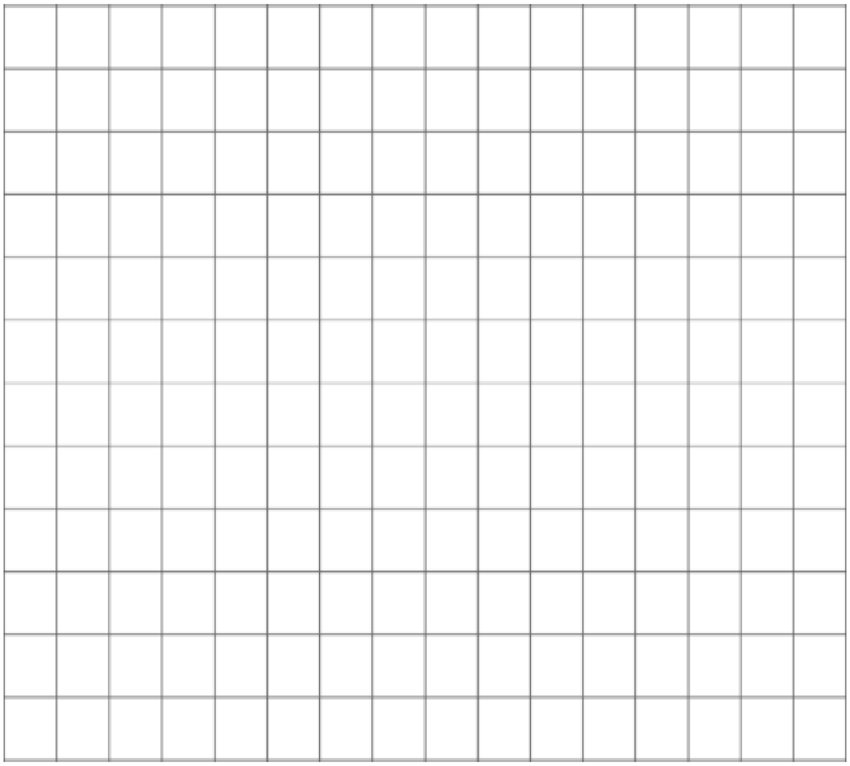
	x-axis	**y-axis**
Scale		

g. On the grid, draw and label coordinate axes with an appropriate scale, and plot the points.

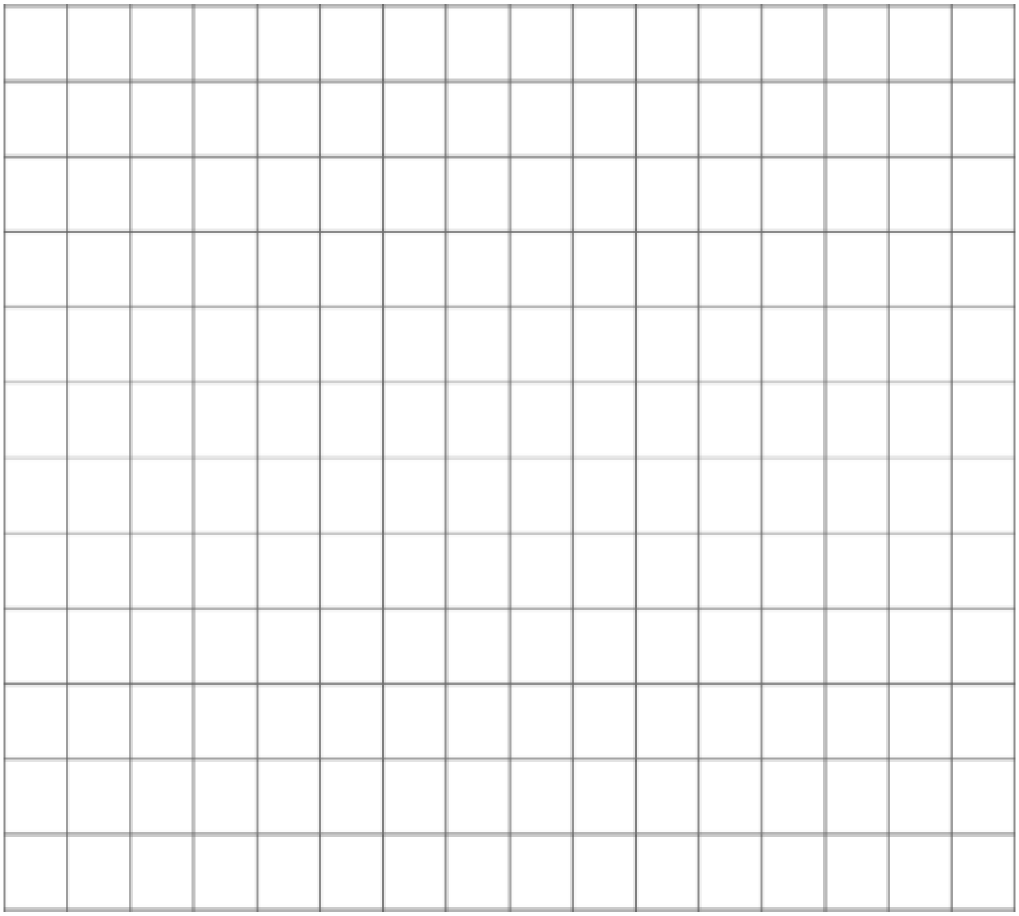
5.2.6 Collaboration: Choosing the Right Scale

1. With your partner, use the given sets of ordered pairs to draw and label coordinate axes with an appropriate scale, and plot the points.

   a. $(50, 50)$, $(0, 0)$, $(-10, -30)$, $(-35, 40)$



b. $\left(\frac{1}{5},\frac{4}{5}\right)$, $\left(\frac{-3}{5},\frac{2}{5}\right)$, $\left(-1\frac{1}{5},\frac{-4}{5}\right)$, $\left(\frac{1}{5},\frac{-3}{5}\right)$



2. Find another group of partners to compare your graphs with. Explain what is the same and different between your graphs and their graphs.

3. If your graphs were different, were both still correct representations of the ordered pairs?

5.2.7 Wrap-Up: Ticket Out the Door

1. When plotting $(-3.5, 40)$ and $(4, -60)$, describe how you would scale the axes.

2. For the points $(-3.5, 40)$ and $(4, -60)$, draw and label coordinate axes with an appropriate scale, and plot the points.